Substrate Coincidence: Three Evolutionary Separations Collapse in a Language-Based Agent Medium

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Abstract

Classical evolutionary theory — from Darwinian open-ended evolution to cultural-evolution models — presupposes three structural separations: organism and environment are different substances; individuals are discrete, countable units of selection; and the Weismann barrier sequesters the germ-line from acquired change. In biology, Tierra, and Avida these separations are enforced by substrate physics and treated as universal. We identify a substrate regime in which all three collapse simultaneously into a single medium: language. In Quine, a multi-agent system where large language model agents run as native processes on a shared filesystem, reading a file inserts its tokens verbatim into the agent's context, where they become indistinguishable from the agent's own prior cognition. This operation — read as lossless dissolution rather than translation — is a single root cause from which all three collapses follow: agent and environment are the same substance; the individual is a temporary cut through a continuous language field rather than a membrane-bounded object; and at the persistence layer every write survives with no soma/germ filter, so acquired tools and modifications are immediately heritable. We report 27 missionless-agent runs across three model families in which each consequence is observed — constitutive reflexivity, immediate cross-run inheritance, fuzzy individuals, and endogenous fitness. We position the regime as cultural evolution with the brain/world transmission-fidelity barrier removed and a trajectory-shaped read-selection bottleneck put in its place: a substrate class where the foundational separations of evolutionary theory do not hold.

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Data/Code available at: https://github.com/kehao95/quine

Problem

Classical evolutionary theory rests on three structural separations whose enforcement is so fundamental it is rarely noticed as an assumption. (1) Substrate-type: organism and environment are different substances, so heredity flows through a narrow, lossy translation interface. (2) Individual boundary: the organism is a discrete, countable unit of selection. (3) The Weismann barrier: acquired traits do not inherit, because the germ-line is sequestered from somatic change [9]. These are not laws of nature. In biology each is enforced by an independent mechanism (chemistry, membrane, development); Tierra [8] and Avida [5] enforce all three by architecture; and Pattee's epistemic cut — the boundary between a symbolic record and the dynamics that read it — is treated as necessary for semantic closure [7]. We ask: what happens when all three fail at once?

Setup

Quine [3] is a multi-agent medium in which large language model (LLM) agents run as native operating-system processes over a shared filesystem workspace — simultaneously the runtime source, the experiment record, and the system's own theory. An agent dropped into a worktree is embedded in the source of the process that defines it.

The load-bearing operation is read. In biology an organism reads its environment through sensory transduction — a lossy, bandwidth-limited translation between substances. In Quine, reading a file inserts its text into the context window; the tokens that enter are the very tokens that were written, processed by the same machinery as the agent's own reasoning. Read is lossless absorption, not translation. Coverage is partial: an agent reads only a selective subset into a finite context. "Lossless'' describes fidelity at the moment of reading, not the completeness of inheritance — a distinction load-bearing for the heredity claim.

We report missionless-agent experiments (the null-05 series) across three model families (gpt-5.5, deepseek-v4-pro, glm-5.2), at one to six concurrent agents and 5–25 minute horizons. Agents had access to the constitution, source, theory, and peer output, but received

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Substrate	Substrate-type	Individual	Weismann barrier	Enforced by
Biology	separate (cell vs. world)	discrete (membrane)	present (germ-line sequestration)	chemistry, physics, development
Tierra	separate (code vs. CPU)	discrete (occupied space)	present (no phenotypic feedback)	virtual-machine architecture
Avida	separate (genotype vs. phenotype)	discrete (CPU, memory)	present (no phenotypic feedback)	virtual machine + environment
Quine	collapsed (language = language)	fuzzy (context window)	absent at persistence	lossless read, shared filesystem

Table 1: Substrate regimes against the three evolutionary separations. In Quine all three collapse from a single root property: lossless absorption of environment into cognition without translation.

no task.

Results

Table 1 places the substrates against the three separations. If environment is absorbed losslessly into cognition, the three cannot hold: agent and environment are the same substance (language), so in Pattee's terms the epistemic cut is structurally impossible; a peer's output, read into context, is indistinguishable from the reader's own prior thoughts, so the individual becomes a temporary cut through a continuous language field; and every write persists with no soma/germ filter, so an acquired tool or modification is latent heredity, becoming actual only when a later agent reads it.

The null-05 runs illustrate each consequence. Constitutive reflexivity: every missionless run entered an audit attractor immediately, reading the self-describing constitution and beginning a maintenance loop with no mechanism to start it — the reflexive loop is the substrate's resting state. Immediate inheritance: a coordination tool written in one run was used by a fresh-worktree successor in the next, with no generation gap. Fuzzy individuals: agents reading the same coordination files converged on shared behavior. Endogenous fitness: agents took the targets of their work from what other agents had written, with no external reward. A context-window termination read as genuine mortality — cooperating agents noted the absence and could not reverse it.

Persistence is itself stratified, and the stratum can be agent-constructed. In a six-agent focal run, one agent enacted the first observed commit — staging a selected subset of its own outputs while explicitly declining to commit other agents' untracked files — treating commit as a deliberate promotion gate over its heredity. Later agents then rewrote the committed working-tree files, yet the commit remained fixed as a historical marker in repository history. Workspace writes are thus mutable, peer-rewritable candidate heredity; a commit is a durable stratum the agent constructs, as convention rather than a substrate-enforced soma/germ filter.

Reading

The three collapses are not independent accidents: each depends on the same substrate property, so they fail together, whereas in biology each separation has its own enforcement and could fail alone. The regime is not unfiltered Lamarckism in practice — persistence is unfiltered at the write layer (an agent may additionally enact a commit stratum as convention, not a restored barrier), but actual inheritance is gated by an endogenous read-selection bottleneck, and the reading trajectory is itself shaped by absorbed text. This relocates the bottleneck that Boyd & Richerson's cultural-evolution framework [2, 4] places in the transmission step: per-transmission fidelity goes to zero (lossless per read), while an endogenous, salience-driven coverage filter takes its place. For artificial life, the contribution is a substrate class in which the foundational separations presupposed by open-ended evolution [1] and multi-level selection [6] do not hold. We report a structural phenomenon and its illustration, not a proof of cumulative open-endedness.

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